Number Theory — problem 6.125

A. Engel — Problem-Solving Strategies · 1998

Each of the positive integers $a_1, \ldots, a_n$ is less than 1951. The least common multiple of any two of these is greater than 1951. Show that

$$\frac{1}{a_1} + \cdots + \frac{1}{a_n} < 2.$$